Bad Blood V2: Attention-Based Population Models and Reference-Conditioned Personalization

Jingyi Guo
Bad Blood Project, University of Notre Dame
Research and Engineering Results Report — October 15, 2026

***Abstract*—Bad Blood V2 combines a compact population convolutional neural network with Squeeze-and-Excitation (CNN+SE) and reference-conditioned PairSE personalization. This 20261015 release adopts personal tail adaptation: the change head, last residual CNN block and its SE gate are fitted separately for each participant. Both photoplethysmogram-only (PPG) and electrocardiogram-plus-PPG (ECG+PPG) versions are retained. Each person's first 80% of ordered segments supplies training and three temporal validation periods; their last 20% is test. Learning rate and update count are selected independently for systolic and diastolic blood pressure (SBP/DBP). Across 30 participants and three source seeds, participant-macro mean absolute error (MAE) is 6.89/4.29 mmHg for PPG and 7.14/4.31 for dual input. Relative to matched additive bias correction, these are reductions of 19.0/20.3% and 21.5/24.4%. Both endpoints improve in the overall mean in every seed. We retain complete controls, individual outcomes, checkpoints and runnable code; the previously examined cohort makes this an exploratory comparison rather than independent external confirmation.**

***Index Terms*—blood pressure, personalization, temporal validation, PPG, ECG, CNN, Squeeze-and-Excitation**

I. Adopted method and scope

The adopted personal method is **tail adaptation with person-specific, endpoint-specific hyperparameters**. Single and dual signals are two input versions of this one method. Head-only fitting and automatic selection between head and tail were evaluated as controls, not retained as alternative deployment policies. The population model and its source training remain unchanged. The reported tail policy includes a validation-selected source checkpoint with zero extra personal updates. This candidate is retained exactly as evaluated; the fitting rules and recorded test outcomes are unchanged in this release.

II. Data and matching

All results use the VitalDB-derived portion of PulseDB [1], not the entire MIMIC-III/VitalDB collection or Notre Dame student recordings. Source CNN+SE training used 1,085 people / 390,600 ten-second windows, validation 192 / 69,120 and main testing 141 / 56,400. Population main-test MAE remains 12.39/8.00 mmHg for ECG+PPG and 12.32/7.85 for PPG-only. These uncalibrated population results are a different task and cohort from personal testing.

Personal evaluation randomly selected 30 of 39 eligible local individual records using seed 20261015, excluding population train/validation/test participants and the four previous personal-development people. The 30 people and their indices are identical to the preceding personal-selection experiment. The first 80% of chronologically ordered eligible segments provides 8,882 development windows. A ten-second temporal guard precedes the final test period, which contains 2,211 windows in total. Splits are not random within a person. Each participant supplies the same development and test data to every control. These target records have been examined in previous experiments; no final-test labels entered this run's fitting or selection, but reuse limits confirmatory interpretation. Earlier six-/sixteen-person, 20%-support results remain in the dated 20261013 archive and are not pooled with this new protocol.

Each input is a filtered, synchronized ten-second waveform at 125 Hz (1,250 samples/channel), min-max scaled independently per channel and segment. ECG then PPG occupy the two input slots; PPG-only zeros the ECG slot throughout source training, reference encoding and inference. PulseDB's ABP-derived scalar SBP/DBP labels provide the targets. No current ABP waveform or current true BP is an inference input. An earlier known scalar BP plus earlier signal is needed for calibration; intermittent cuff labels are format-compatible but constitute a separate, much sparser evaluation setting.

III. Population architecture and training

A. Signal representation and network

The supplied filtered ECG and PPG fields are scaled independently within each window to the unit interval. The tensor shape is $N \times 2 \times 1250$, ordered ECG then PPG. The PPG-only fit is trained from scratch without ECG. An internal mask zeros ECG in both ordinary forward inference and reference encoding. Keeping the two-slot representation maintains a controlled parameter-matched comparison; it is not a claim that ECG is needed for the PPG model.

Each residual block has a kernel 7 convolution, BN and SiLU, followed by a kernel 5 convolution and BN. A stride-matched kernel 1 projection and BN form the skip branch; SiLU follows the sum. The SE mechanism [2] reweights learned feature channels after the final residual stage. These channels mix signal features; their weights are not direct sensor importance scores. CBAM and temporal-only attention were also compared using the same backbone. Adding SE increases parameters from 37,666 to 38,296, about 1.7%.

B. Objective, optimizer and checkpoint selection

For normalization $N(\cdot)$, the physical-unit population prediction is

$$p(X) = \mu + \sigma \odot f_\theta(N(X)), \tag{1}$$

TABLE I
CNN+SE ARCHITECTURE; SHAPES OMIT BATCH DIMENSION.

Stage	Operation and output
Input	2 channels × 1,250 samples; no initial decimation
Stem	Conv1d 2→8, kernel 7, stride 2, pad 3; BN; SiLU. 8×625
Block1	Residual 8→16, stride 2. 16×313
Block2	Residual 16→32, stride 2. 32×157
Block3	Residual 32→48, stride 2. 48×79
SE gate	Temporal mean; Linear 48→6; ReLU; Linear 6→48; sigmoid; multiply channels
Readout	Global temporal mean; dropout 0.15; Linear 48→32; SiLU; Linear 32→2

TABLE II
POPULATION FITS AND VALIDATION-SELECTED CHECKPOINTS. BOLD: LOWEST VALIDATION SCORE PER INPUT CONFIGURATION.

Input	Seed	Run epochs	Best epoch	Val. score
dual	42	9	1	**10.291**
dual	43	9	1	10.457
dual	44	9	1	10.421
ppg	42	9	1	**10.374**
ppg	43	9	1	10.583
ppg	44	9	1	10.406

TABLE III
DUAL-INPUT ARCHITECTURES: THREE-SEED PARTICIPANT-MACRO MAE (SBP / DBP), MMHG. BOLD: LOWEST PER PRESSURE AND SPLIT.

Model	Parameters	Validation	Main test
Plain CNN	37666	**12.41** / 8.33	12.59 / 8.36
CNN+SE	38296	12.51 / **8.26**	**12.39** / **8.00**
CNN+temporal	37680	12.49 / 8.32	12.58 / 8.31
CNN+CBAM	38310	12.52 / 8.29	12.42 / 8.11
CNN+additive	38296	12.44 / 8.34	12.66 / 8.18
CNN+SE-all	38670	12.47 / 8.32	12.49 / 8.23

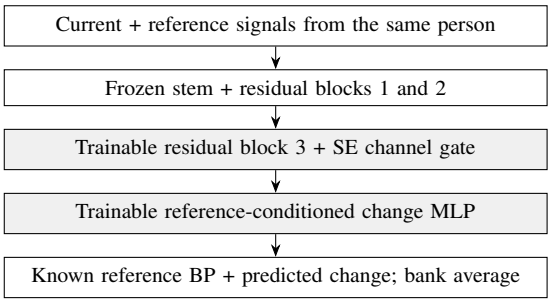


Fig. 1. Adopted personal adaptation. Gray components update on one person's data; early reference and current branches share the same weights.

where training-only label center $\mu = (115.41765, 62.92810)$ and scale $\sigma = (18.94188, 12.08322)$ are stored in the checkpoint. The objective is mean squared error in standardized SBP/DBP, not a mixture of test metrics. Both modes use AdamW, learning rate 0.001, weight decay 0.001, batch 256, input noise SD 0.003 and gradient-norm clipping at 5. A plateau scheduler halves the learning rate after three validation non-improvements. Maximum training is 40 epochs, with early-stopping patience 8. Checkpoints minimize the mean of SBP and DBP participant-macro validation MAE. All six SE fits stopped after nine epochs and retained epoch 1 (Table II); later epochs were evaluated but did not improve that criterion. Seeds 42,43,44 quantify training variation. Seed 42 is the representative within each modality because its validation score is best, not because of a test-set ranking.

IV. REFERENCE-CONDITIONED ARCHITECTURE

PairSE shares one CNN+SE encoder between an earlier reference waveform and a currently observed waveform. Its 48-dimensional features and two standardized population predictions are denoted h_r, p_r and h_c, p_c. With known reference BP y_r, it constructs

$$z_{cr} = [h_c - h_r, h_r, y_r, p_c - p_r], \tag{2}$$

$$z_{rr} = [0, h_r, y_r, 0], \tag{3}$$

$$\widehat{y}_{c|r} = y_r + (p_c - p_r) + g_\phi(z_{cr}) - g_\phi(z_{rr}). \tag{4}$$

The pressure-change multilayer perceptron (MLP) is 100–64–32–2 with Sigmoid Linear Unit (SiLU) activations and 8,610 parameters. Full PairSE has 46,906 parameters. Identity subtraction recovers the known BP exactly when both inputs are identical. Final prediction averages over every known reference in that person's development bank. Features and BP use distinct, source-training-only normalization statistics. This is current-pressure estimation using observed current signals, not prediction of an unseen future waveform.

The design adapts paired-reference and transfer-learning ideas [3], [4], [5], rather than reproducing their published networks. We combine a compact 1D residual encoder, a final-stage SE gate, explicit reference BP, identity anchoring and all-reference averaging. Group pair-head training freezes the encoder and trains on the same 390,600 source windows with random non-self references from the same participant. It uses Adam, learning rate 0.001, batch 2,048, standardized mean squared error (MSE), gradient clipping at 5, at most 20 epochs and patience 5. Source validation selects checkpoints; target-person data is absent from this group training.

V. PERSONAL TAIL ADAPTATION: EXACT PIPELINE

A. Trainable and frozen parameters

The trainable components are `change.*`, `encoder.body.6.*` and `encoder.gates.6.*`: the change head, last residual block and final SE gate, totaling **33,336 parameters**. Earlier encoder blocks, population readout weights, feature/BP normalization and BatchNorm running statistics remain fixed. BatchNorm affine scale/bias inside the last block can change. Dropout is disabled during personal fitting. The frozen population readout can still produce different values because its input features are adapted.

B. Only the person's first 80% selects their settings

For each person and source seed, create three expanding temporal folds within their development bank: train 0–50%, validate 50–65%; train 0–65%, validate 65–80%; train 0–80%,

validate 80–100%. These percentages are of the *development bank*, not the complete recording. Validation starts at least 10 seconds after the latest training window ends. The last 20% of the full record is never used in these folds. Candidate fitting always restarts from the same source PairSE checkpoint.

For each endpoint separately, evaluate learning rates $\{10^{-4}, 3 \times 10^{-4}\}$ and update counts $\{10, 25, 50, 100\}$, plus the source model with zero personal updates. Minimize the equally weighted average of the three validation MAEs. Ties prefer fewer updates, then smaller learning rate. SBP and DBP may select different settings; another target person's data has no role in that choice. After selection, restart from source and refit on the entire development bank. There are two saved endpoint copies per participant/source seed. Predictions take SBP from its selected copy and DBP from its selected copy, not both outputs from an arbitrarily chosen checkpoint.

Every optimizer step uses all valid chronological reference/current pairs in that fold's training bank, requiring reference end + 10 seconds $\leq$ current start. The endpoint-specific objective is standardized MSE; the optimizer is Adam, with no weight decay and gradient norm capped at 5. Update counts are optimizer steps, not epochs. Pairing reuses observed labels and creates no new independent observations. Frozen early features are cached; exact accumulated pair gradients propagate through the trainable tail. Final testing freezes all weights and uses only known development references and later signal inputs.

C. Controls and source seeds

Personal mean predicts the mean development BP. Additive bias correction is $p(X_c) + |S|^{-1} \sum_{r \in S} [y_r - p(X_r)]$, using the same modality's population model. It already tracks waveform-dependent changes, unlike the constant mean. Other matched controls are the old fixed head-only PairSE (learning rate 0.0003, 25 joint-endpoint updates), previous single-period personal selection, three-fold endpoint-specific head-only fitting, and validation-selected head/tail scope. Only the tail candidate group is the adopted method.

Seeds 42/43/44 change the population training initialization, shuffled training order and stochastic source pair-head training, yielding different initializers. They do not change participants, temporal splits or reference budgets. Personal fitting uses all valid pairs, rather than a newly randomized subset. Three-seed averages are averages of separate evaluated runs, not an ensemble.

VI. Metrics and overall results

We compute MAE, root MSE (RMSE), signed Bias, sample error standard deviation (SD, denominator $n-1$), and fractions within 5, 10 and 15 mmHg, separately for SBP/DBP. Participant-macro metrics give each of the 30 people equal weight; pooled metrics give each test segment equal weight. Macro SD averages each person's error SD and differs from pooled SD. All tables average three seeds. Best values are bold within each table and endpoint; Bias is ranked by distance from zero. Full per-person and per-seed metrics are available in the release.

A. PPG only

Tail adaptation reaches 6.89 / 4.29 mmHg MAE. Relative to personal mean, this is a reduction of 5.35 / 1.73 mmHg (43.70 / 28.77%). Relative to additive bias correction, the reduction is 1.62 / 1.09 mmHg (19.00 / 20.34%). Both endpoints improve in the overall mean in each of the three seeds.

TABLE IV
PPG ONLY: PARTICIPANT-MACRO RESULTS. SBP / DBP.

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.24 / 6.02	8.51 / 5.38	**6.89 / 4.29**
RMSE	14.32 / 7.11	10.12 / 6.17	**8.51 / 5.16**
Signed Bias	-6.07 / -2.80	-2.81 / -1.48	**-0.70 / -0.74**
Sample error SD	9.13 / 4.69	7.12 / 3.89	**6.52 / 3.68**
≤ 5 (%)	31.14 / 58.60	42.98 / 59.35	**47.14 / 65.91**
≤ 10 (%)	56.26 / 81.76	70.29 / 86.43	**75.87 / 92.65**
≤ 15 (%)	72.42 / 89.22	85.07 / 94.18	**90.50 / 98.91**

B. ECG+PPG

Tail adaptation reaches 7.14 / 4.31 mmHg MAE. Relative to personal mean, this is a reduction of 5.11 / 1.71 mmHg (41.71 / 28.37%). Relative to additive bias correction, the reduction is 1.96 / 1.39 mmHg (21.54 / 24.37%). Both endpoints improve in the overall mean in each of the three seeds.

TABLE V
ECG+PPG: PARTICIPANT-MACRO RESULTS. SBP / DBP.

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.24 / 6.02	9.09 / 5.70	**7.14 / 4.31**
RMSE	14.32 / 7.11	10.86 / 6.54	**8.86 / 5.22**
Signed Bias	-6.07 / -2.80	-2.54 / -1.26	**-1.57 / -0.77**
Sample error SD	9.13 / 4.69	8.11 / 4.52	**7.18 / 4.00**
≤ 5 (%)	31.14 / 58.60	39.95 / 55.02	**47.01 / 66.76**
≤ 10 (%)	56.26 / 81.76	66.51 / 84.10	**74.28 / 92.23**
≤ 15 (%)	72.42 / 89.22	82.24 / 94.19	**88.56 / 98.42**

C. What the tail and scope ablations show

Compared with head-only adaptation, tail adaptation reduces macro MAE by 0.70/0.25 mmHg for PPG and 1.25/0.58 for dual input. PPG tail MAE varies across seeds with SD 0.36/0.13 mmHg; dual tail SD is 0.11/0.11. PPG has slightly lower mean error; dual has more consistent source-seed results. A paired participant bootstrap interval for the PPG-minus-dual MAE difference crosses zero for both endpoints, so these results do not establish a universal modality winner.

Automatic scope selection gives 7.15/4.23 (PPG) and 7.23/4.52 (dual). Tail-only selection gives 6.89/4.29 and 7.14/4.31 respectively. Scope selection therefore slightly improves PPG DBP but worsens PPG SBP and both dual endpoints. The adopted policy fixes the trainable scope to the tail while retaining personal selection of learning rate and update count. This release decision follows the exploratory comparison, not an untouched external-test selection claim.

D. Individual outcomes and uncertainty

After averaging each person's errors across seeds, 10/30 PPG participants and 13/30 dual participants beat both controls on both pressures simultaneously. The strict count in each seed is 11/9/7 (PPG) and 12/12/15 (dual); 3 and 10 people, respectively, win in every seed. These are secondary individual counts, distinct from the overall mean, which improves in all six modality/seed runs. Some large gains influence the mean. Excluding the largest beneficiary as a descriptive sensitivity check still leaves tail-minus-bias MAE differences of -0.85/-0.55 mmHg (PPG) and -1.33/-1.00 (dual). Official results retain all people. The full report tables and machine-readable results preserve unfavorable individual outcomes rather than claiming a benefit for every person.

VII. Implementation, artifacts and verification

The current folder is `final_results_20261015`. The adopted method is `rolling_tail`. Six source PairSE initializers and 360 personal endpoint checkpoints ($30 \times 3 \times 2 \times 2$) are provided, with all selections and checksums. A portable fitting command accepts only one person's development signals, known BP and timestamps. The inference command accepts current signals, earlier references and the two endpoint checkpoints, never current query BP. Raw reference signals are not redistributed; follow the frozen manifest to recover the public source files. A personal checkpoint is not transferable to another person without a fresh personal fit.

All 180 person/modality/seed units and 1,080 internal validation folds completed. Independent verification replayed 720 endpoint models with maximum prediction difference 1.53×10^{-5} mmHg and confirmed matched budgets. The completed record hashes match the archived summary and verification. This establishes software/result consistency, not clinical certification. The earlier dated V2 results and V1 are preserved. The current portable release is additionally checked against archived endpoint predictions and published with its own manifest.

VIII. Conclusion

V2 retains both population CNN+SE input modes and adopts reference-conditioned tail personalization with independent within-person selection for each pressure. The controlled 30-person comparison shows consistent overall MAE improvement over personal mean and additive bias correction across all three seeds for both modalities. Tail adaptation improves the overall result beyond head-only fitting. The next independent evaluation should preserve this policy and assess new people, including the distinct low-cost-sensor and cuff-reference setting at Notre Dame.

References

[1] W. Wang et al., "PulseDB: A large, cleaned dataset based on MIMIC-III and VitalDB for benchmarking cuff-less blood pressure estimation methods," Frontiers in Digital Health, vol. 4, 1090854, 2023. doi:10.3389/fdgth.2022.1090854.

[2] J. Hu, L. Shen, and G. Sun, "Squeeze-and-Excitation Networks," CVPR, pp. 7132–7141, 2018.

[3] B. K. Mekonnen et al., "Accurate and 30-plus days reliable cuffless blood pressure measurements with 9-minutes personal photoplethysmograph data and mixed deduction learning," Scientific Reports, vol. 14, 23722, 2024. doi:10.1038/s41598-024-75583-y; author correction doi:10.1038/s41598-024-79326-x.

[4] O. Schlesinger et al., "Estimation and Tracking of Blood Pressure Using Routinely Acquired Photoplethysmographic Signals and Deep Neural Networks," Critical Care Explorations, vol. 2, e0095, 2020.

[5] J. Leitner, P.-H. Chiang, and S. Dey, "Personalized Blood Pressure Estimation Using Photoplethysmography: A Transfer Learning Approach," IEEE JBHI, vol. 26, pp. 218–228, 2022.

COMPLETE MATCHED COMPARISONS: PPG ONLY

TABLE VI
PPG ONLY: ALL PREDECLARED METHODS, PARTICIPANT-MACRO METRICS. SBP / DBP.

Method	MAE	RMSE	Bias	Error SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
Population (no calibration)	14.66 / 10.60	16.16 / 11.32	-2.11 / **0.14**	7.12 / 3.89	21.21 / 27.95	42.39 / 53.03	58.84 / 73.64
Personal mean	12.24 / 6.02	14.32 / 7.11	-6.07 / -2.80	9.13 / 4.69	31.14 / 58.60	56.26 / 81.76	72.42 / 89.22
Bias correction	8.51 / 5.38	10.12 / 6.17	-2.81 / -1.48	7.12 / 3.89	42.98 / 59.35	70.29 / 86.43	85.07 / 94.18
Fixed PairSE	7.75 / 4.80	9.43 / 5.76	-3.16 / -1.48	6.92 / 3.93	44.81 / 61.47	71.87 / 89.12	86.99 / 97.59
Single-period selection	7.72 / 4.67	9.53 / 5.67	-1.97 / -0.82	6.96 / 3.95	46.28 / 64.75	72.89 / 89.32	86.10 / 96.58
Head-only adaptation	7.59 / 4.54	9.39 / 5.51	-2.14 / -1.18	6.98 / 3.88	45.36 / 65.42	73.42 / 90.48	87.42 / 97.43
Tail adaptation (adopted)	**6.89** / 4.29	**8.51** / **5.16**	**-0.70** / -0.74	**6.52** / **3.68**	**47.14** / 65.91	**75.87** / **92.65**	**90.50** / **98.91**
Scope selection (comparison)	7.15 / **4.23**	8.80 / 5.17	-1.15 / -0.91	6.54 / 3.75	46.42 / **68.18**	75.25 / 92.50	89.24 / 98.26

TABLE VII
PPG ONLY: SEED-LEVEL PARTICIPANT-MACRO MAE; EACH COLUMN USES THE SAME 30 PEOPLE.

Method	Seed 42	Seed 43	Seed 44
Population (no calibration)	14.92 / 10.48	14.83 / 10.65	14.23 / 10.65
Personal mean	12.24 / 6.02	12.24 / 6.02	12.24 / 6.02
Bias correction	8.45 / 5.37	8.58 / 5.32	8.50 / 5.45
Fixed PairSE	7.78 / 4.76	8.00 / 4.93	7.47 / 4.70
Single-period selection	7.59 / 4.64	8.13 / 4.94	7.44 / 4.45
Head-only adaptation	7.84 / 4.45	7.81 / 4.94	7.13 / 4.23
Tail adaptation (adopted)	7.31 / 4.43	6.68 / 4.25	6.69 / 4.18
Scope selection (comparison)	7.41 / 4.32	7.22 / 4.29	6.82 / 4.10

TABLE VIII
PPG ONLY: POOLED SEGMENT-WEIGHTED CONTROLS AND ADOPTED MODEL, SBP / DBP.

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.33 / 5.93	9.27 / 5.55	**8.05** / **4.31**
RMSE	16.04 / 8.03	12.16 / 7.25	**10.45** / **5.50**
Signed Bias	-3.52 / -2.29	**-1.34** / -1.17	1.43 / **0.44**
Sample error SD	15.65 / 7.69	12.08 / 7.15	**10.33** / **5.48**
$\leq$ 5 (%)	26.10 / 53.82	35.46 / 56.25	**40.84** / **66.15**
$\leq$ 10 (%)	50.93 / 82.68	63.00 / 84.11	**68.42** / **93.03**
$\leq$ 15 (%)	67.21 / 92.76	81.71 / 95.19	**85.71** / **98.81**

COMPLETE MATCHED COMPARISONS: ECG+PPG

TABLE IX
ECG+PPG: ALL PREDECLARED METHODS, PARTICIPANT-MACRO METRICS. SBP / DBP.

Method	MAE	RMSE	Bias	Error SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
Population (no calibration)	14.80 / 10.35	16.48 / 11.21	**-0.86** / 1.10	8.11 / 4.52	23.21 / 28.24	42.64 / 51.07	58.06 / 74.94
Personal mean	12.24 / 6.02	14.32 / 7.11	-6.07 / -2.80	9.13 / 4.69	31.14 / 58.60	56.26 / 81.76	72.42 / 89.22
Bias correction	9.09 / 5.70	10.86 / 6.54	-2.54 / -1.26	8.11 / 4.52	39.95 / 55.02	66.51 / 84.10	82.24 / 94.19
Fixed PairSE	8.78 / 5.05	10.52 / 6.04	-3.30 / -1.42	7.82 / 4.27	37.88 / 59.45	67.19 / 88.46	83.34 / 96.09
Single-period selection	8.17 / 4.78	9.99 / 5.76	-2.43 / **-0.69**	7.65 / 4.20	43.48 / 62.90	70.51 / 88.73	84.11 / 96.55
Head-only adaptation	8.39 / 4.89	10.13 / 5.90	-2.89 / -1.06	7.44 / 4.27	41.79 / 61.77	69.25 / 87.37	83.79 / 96.46
Tail adaptation (adopted)	**7.14** / **4.31**	**8.86** / **5.22**	-1.57 / -0.77	7.18 / **4.00**	**47.01** / **66.76**	**74.28** / **92.23**	**88.56** / **98.42**
Scope selection (comparison)	7.23 / 4.52	8.94 / 5.51	-1.86 / -0.78	**7.13** / 4.17	46.91 / 64.83	73.99 / 90.71	88.02 / 97.50

TABLE X
ECG+PPG: SEED-LEVEL PARTICIPANT-MACRO MAE; EACH COLUMN USES THE SAME 30 PEOPLE.

Method	Seed 42	Seed 43	Seed 44
Population (no calibration)	14.52 / 9.84	14.82 / 10.49	15.06 / 10.72
Personal mean	12.24 / 6.02	12.24 / 6.02	12.24 / 6.02
Bias correction	9.01 / 5.65	8.88 / 5.55	9.39 / 5.90
Fixed PairSE	8.55 / 5.10	8.69 / 5.00	9.09 / 5.05
Single-period selection	7.80 / 4.66	8.74 / 5.12	7.96 / 4.56
Head-only adaptation	7.94 / 4.71	8.95 / 4.85	8.28 / 5.12
Tail adaptation (adopted)	7.16 / 4.44	7.23 / 4.26	7.01 / 4.23
Scope selection (comparison)	7.38 / 4.58	7.38 / 4.44	6.94 / 4.53

TABLE XI
ECG+PPG: POOLED SEGMENT-WEIGHTED CONTROLS AND ADOPTED MODEL, SBP / DBP.

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.33 / 5.93	10.38 / 6.00	**7.98** / **4.24**
RMSE	16.04 / 8.03	13.41 / 7.71	**10.33** / **5.50**
Signed Bias	-3.52 / -2.29	-1.47 / -1.37	**0.60** / **0.37**
Sample error SD	15.65 / 7.69	13.32 / 7.58	**10.30** / **5.48**
$\leq$ 5 (%)	26.10 / 53.82	31.31 / 51.68	**41.28** / **67.75**
$\leq$ 10 (%)	50.93 / 82.68	57.14 / 79.98	**69.24** / **92.67**
$\leq$ 15 (%)	67.21 / 92.76	77.07 / 94.86	**85.36** / **98.49**